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Special Collection: Wave-Structure Interaction

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ABSTRACT

The misalignment between flow and rotor can significantly alter the efficiency of the tidal stream turbine (TST), and therefore, it is vital to predict the flow in the tidal field and the performance of the TST under yaw-offset conditions. First, this paper implements a high-precision Lagrangian dynamic sub-grid-scale model based on the large-eddy simulation (LES) method. A classical computational fluid dynamics benchmark case is selected to validate the accuracy of the dynamic LES (DLES) method. The results indicate that the newly implemented dynamic LES method exhibits reduced dissipation and effectively captures the local effects of non-uniform flow fields, including vortex structures. Second, an efficient high-fidelity numerical method (AL-DLES) for forecasting the TST wake is presented by integrating an actuator line (AL) code with the aforementioned DLES method based on the Lagrangian framework. After comparing the experimental results, it was discovered that the newly developed AL-DLES coupling approach, which addresses the issues of challenging turbine meshing, rapid wake dissipation, and insufficient flow field fidelity in previous methods, can accurately simulate the forces acting on the TST while also capturing detailed characteristics of the flow field. Furthermore, the study will be extended to investigate the TST wake dynamics under various yaw-offset conditions, exploring the mechanisms of instability evolution in wake meandering. Meanwhile, the latest third-generation (Ω_{new}) vortex identification program is implemented and successfully applied to the wake vortex visualization of the TST under yaw-offset conditions. Through a comparative analysis of three distinct vortex identification approaches, it was demonstrated that the Ω_{new} method exhibits superior accuracy in capturing the vortex system located behind the rotor, eliminating the need for manual threshold selection. In addition, it is capable of simultaneously capturing both strong and weak vortices, which is a vital aspect for future wake research.

I. INTRODUCTION

Since the turn of the 21st century, several nations have started focusing on marine renewable energy in response to the increasing depletion of conventional energy sources and environmental damage.[1–3] Renewable tidal energy's power density and predictability are both high compared to other resources.[4–6] A typical device that utilizes tidal energy to generate electricity is the horizontal-axis tidal stream turbine (TST).[7,8] The turbine's rotating disk is set perpendicular to the water flow, and the torque generated by the blade under the action of the water flow drives the turbine to revolve around the main shaft, thus converting the water's kinetic energy into rotating mechanical energy. Then, the generator, driven by the nacelle, main shaft, and transmission system, generates electricity. The TST self-starts and is highly efficient and stable using the three distributed blades' arrangement.[9,10]

Tidal current is the product of periodic horizontal movements of the seawater, so its direction varies as well periodically. This has effects not only on fixed-to-bed but also on floating turbine concepts. For

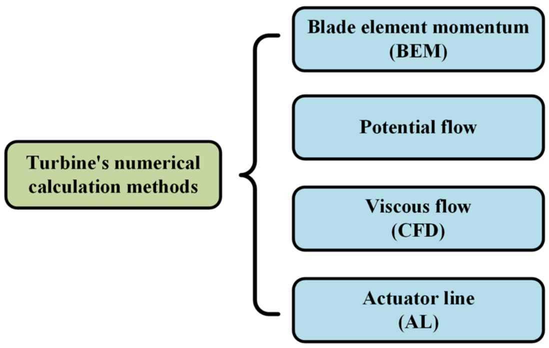

FIG. 1. Different numerical calculation methods for tidal turbines.

instance, the movement of floating carriers can also cause a certain deviation between the rotation axis of the TST and the direction of the tidal current. Likewise, the rotating axis of the TST is sometimes misaligned due to an inability or low response to track the flow direction, leading to a yaw state.[11] In other words, there is no strictly directional reciprocating flow in sea conditions. The yaw state reduces the output power of the TST,[12] so it is useful for adjusting the turbine load at a high-speed ratio. However, it causes blade load fluctuations, resulting in dynamic fatigue load and affecting the blade's lifespan.[13]

Currently, there are four different types of numerical calculation methods for the turbine,[14–16] illustrated in Fig. 1.

A blade element momentum (BEM) numerical method is easy to program and provides fast, reliable performance predictions of the turbine in a steady state.[17] Some scholars[18,19] employ the steady BEM method to calculate the load characteristics of the TST (that is, under quasi-steady conditions), whilst others[20–22] introduce linear wave theory to get insights into the impact of the wave-induced load. Currently, the BEM method is unable to predict the transient loads on the blades or analyze the three-dimensional flow field characteristics, limiting its potential for turbulence-related problems and multiple wake interactions.[14]

The potential flow method has the benefit of simulating the three-dimensional flow around the blades as well as being able to calculate the unsteady hydrodynamic loads of the TST under yaw-offset conditions. However, this method is unable to account for the effect of fluid viscosity on the flow field or the occurrence of flow separation, and the accuracy of the turbine load prediction diminishes at low-speed ratios.[23] In addition, the calculation is also vulnerable to divergence when flow separation and dynamic stall phenomena occur.[24] The potential flow method's application in the TST field still needs further development.[14]

In recent decades, with the improvement of computer performance, the viscous flow computational fluid dynamics (CFD) method has been extensively used in the TST research, mainly focusing on blade optimization,[25–27] diffuser augmented design,[28,29] turbine cavitation,[30,31] vortex generators,[32–34] blocking effect,[35,36] column effect,[37,38] turbulence intensity,[39,40] wave effect,[41–43] and array interference[44–46]. The research findings indicate that altering these parameters may improve or impair the turbine's hydrodynamic characteristic. The aforementioned studies place a greater emphasis on the TST's hydrodynamic characteristic in axial-flow conditions, lacking discussion on the mechanism of wake evolution and its impact on energy conversion efficiency.

The above CFD method requires significant computational resources. Sørensen and Shen[47] proposed the actuator line (AL) method in the field of wind energy in order to find a workable compromise between low-order models and three-dimensional simulations of the real geometry, but it has then been used in the field of tidal energy.[48–50] The proposed methodology incorporates a body-force scheme into a CFD solver. The AL method involves the substitution of blades with rotating lines, upon which a force distribution is applied. The determination of this is accomplished by utilizing tabulated airfoil data. Due to the elimination of turbine geometry, the necessity for boundary layer resolution is obviated, resulting in a sparser grid and significantly reduced computational requirements. Moreover, if a tidal array farm is considered, the prediction of the wake interaction through an AL method significantly simplifies the computational complexity and can realize the fine numerical simulation of the TST wake with a wide range of spatiotemporal evolution.

Some academics have recently employed the AL method for numerical analysis. Lin *et al.*[51] compared in detail the differences between the viscous CFD and AL methods in turbine wake simulation under axial-flow conditions and found that the two methods captured similar wake characteristics. However, the CFD method requires body-fitted mesh for turbine blades, which is not only complicated in grid division, but also significantly longer in calculation time than the AL method. Baba-Ahmadi and Dong[52] and Apsley *et al.*[53] combined the AL and large-eddy simulation (LES) methods to calculate the hydrodynamic performance of a single TST and multiple TST arrays under axial-flow conditions. The findings demonstrated that this method can accurately predict the average and turbulent characteristics of the flow field and track the evolution of vortex systems (tip vortex, root vortex, etc.). The primary focus of Krogstad and Adaramola[54] was to study the power output and wake dynamics of a turbine at various inflow velocities and yaw angles. They noticed that as the yaw angle increased, the wake became narrower and deflected to one side. Sørensen *et al.*[55] and Qian *et al.*[13] studied the wake dynamics of the TST in axial and yawed states considering the nacelle obstruction effect. In summary, the hydrodynamic behavior and wake structure of the TST have been comprehensively investigated under axial-flow conditions, with particular emphasis on dominant flow structures,[56] the stability of the vortex system,[57] and wake meandering.[58–60] Despite several benefits, the application of the AL method in the reference has been limited to verification under yaw-offset conditions.[61] There is relatively little research on the performance and wake structure of the TST across a wide range of yaw angles. Despite these emerging applications, accurately predicting velocity losses in the wake[62,63] and turbine wake deviation[64–66] remains challenging. Therefore, further research is necessary to achieve a more comprehensive understanding and clarify the strengths and limitations of this approach in yaw states, which is crucial for enhancing the output power of tidal array farms.[67]

The primary objective of this research is to examine the performance and wake configuration of the TST subjected to yaw-offset conditions, which are crucial to achieving reliable tidal array farm design. High-fidelity CFD simulations are conducted using the in-house solver (JUST-FOAM-Turbine), which is based on the AL-DLES coupling methodology.[68] This incompressible solver is designed to analyze the dynamic characteristics of turbine wakes. The JUST-FOAM-Turbine solver was developed using the open-source code library OpenFOAM. The turbine employs the AL code, while the wake simulation utilizes the dynamic LES (DLES) method based on a Lagrangian framework. First, the reliability of the solver is validated against experimental results. Second, the hydrodynamic performance and the mechanisms of instability evolution in the TST's wake meandering under varying yaw misalignment are investigated.

Different from previous studies, the novelty of the current paper lies in proposing a high-precision AL-DLES method that simultaneously integrates actuator line (AL), dynamic LES (DLES), and a new-generation vortex identification technique. This method addresses the issues of insufficient fidelity in the flow field and rapid wake dissipation encountered in earlier research. Furthermore, the internal causes of output power fluctuations resulting from yaw misalignment are analyzed, a factor that has not been quantified in prior studies. The subsequent sections of the paper will be structured as follows. Section II presents a comprehensive overview of the computational model, encompassing dynamic LES governing equations, the employed actuator line model, and vortex identification methods. In the discussions section, after validating the current solver, the results of the yawed cases are evaluated. Finally, the conclusions section is presented.

II. NUMERICAL METHODS

A. Dynamic large-eddy simulation

Within the framework of the LES, turbulent structures may be classified into two distinct categories, namely, large-scale (resolved) and small-scale (unresolved) eddies, which are differentiated by a process of filtering. Spatial filtering is used to solve resolved-scale turbulent structures, while the sub-grid-scale (SGS) model is utilized to simulate unresolved-scale turbulent structures. For the incompressible LES, the grid-filtered continuity and Navier–Stokes (N-S) equations are formulated as follows:[69,70]

$$\frac{\partial \overline{u}_i}{\partial x_i} = 0, \tag{1}$$

$$\frac{\partial \overline{u}_i}{\partial t} + \frac{\partial}{\partial x_j}(\overline{u}_i \overline{u}_j) = -\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i} - \frac{\partial \tau_{ij}}{\partial x_j} + \frac{\mu}{\rho}\frac{\partial^2 \overline{u}_i}{\partial x_j \partial x_j} + f_i, \tag{2}$$

where the overbar represents grid-scale filtered variables. u_i stands for the velocity's Cartesian component in the i direction and ρ is the water density. μ is the dynamic viscosity and p denotes pressure that ignores the Boussinesq buoyancy effect. The N-S equation's terms, which are arranged from left to right, denote transient, convection, pressure, SGS stress tensor, molecular viscous stress, and the external body force f_i from the AL model acting on the TST, respectively. There are unresolved-scale effects in the SGS stress tensor, as shown in Eq. (3), which is essential for closing Eq. (2)

$$\tau_{ij} = \overline{u_i u_j} - \overline{u}_i \overline{u}_j. \tag{3}$$

Constructing the SGS model has consistently been a pivotal concern in LES research. The SGS model's purpose is to resolve the SGS stress tensor τ_{ij} specified in Eq. (3). The accuracy and applications of various SGS models vary for different physics issues. Currently, there are two categories of SGS models: direct solving (also known as large-eddy simulation, or LES), and dynamic solving (also known as dynamic large-eddy simulation, or DLES).[71–73]

(1) The coefficients in directly solved SGS models are constants, and the most typical one in this category is the standard Smagorinsky model with constant coefficients.[74]

(2) The coefficients in the dynamically solved SGS models are computed based on the real circumstances of the local flow field, avoiding the randomness of artificially given coefficients for various flow problems. This class of SGS model has been successively proposed and gradually developed by academics like Germano *et al.*,[75] Lilly,[76] Ghosal *et al.*,[77] and Meneveau *et al.*,[78] primarily including the dynamic Smagorinsky SGS model, the local dynamic Smagorinsky SGS model, the new Lagrangian dynamic Smagorinsky SGS model, and so on.

In this study, the LES coupled with a novel Lagrangian dynamic Smagorinsky SGS model[72] is integrated into OpenFOAM, referred to as dynamic LES (DLES),[73] for turbulence modeling. The DLES approach, featuring the Lagrangian dynamic Smagorinsky SGS model, takes into account both local and historical influences, offering advantages for complex flows characterized by local non-equilibrium turbulence compared to the conventional LES method (Standard Smagorinsky SGS model). It improves computational precision in wall-adjacent regions[79–81] and effectively captures the helical blade vortex structures, including those at the blade tip and root within the wake of the TST.[69,72] In addition, the vanDriest filter delta[82] is employed within the DLES framework for wall treatment, aimed at enhancing the predictive accuracy for the vicinity of the wall.

The present investigation employs the collocated finite volume method (CFVM) for discretizing the governing equations, specifically Eqs. (1) and (2). The momentum equation is addressed through Issa's transient PISO algorithm,[83] which has limitations in resolving all coupled equations either iteratively or in a coupled fashion. Rather, this approach separates the non-linear velocity–pressure system into an implicit predictor followed by several explicit correction phases. In addition, the second-order QUICK scheme is used to reconstruct the convective phase flux. The time-stepping process employs the second-order bounded semi-implicit Crank–Nicholson (CN 1.0) method, while the remaining terms utilize the second-order unbounded Gaussian linear scheme based on the central difference approach. To resolve the linear system associated with velocity and turbulent kinetic energy (TKE), the PBiCG solver is implemented. Conversely, the linear system related to pressure is addressed with the PCG solver, supported by the GAMG preconditioner.

B. Actuator line model

The AL model, initially introduced by Sørensen and Shen,[47] represents a highly effective methodology that integrates conventional blade element momentum theory with the Navier–Stokes equations to accurately analyze the primary dynamics of the wake surrounding the TST. This AL approach incorporates several actuator lines rotating around the axis of each blade, tailored to the specific count of TST blades. As illustrated in Fig. 2, a set of actuator points (APs) is positioned at the midpoint of each actuator element (AE) along the corresponding actuator line (AL). The influence of the TST on the flow field is simulated by loading the body force at the AP position, subsequent to which the performance and wake dynamics of the TST are evaluated by solving the N–S equations. Consequently, it is not necessary to solve the blade boundary layer, which means that three-dimensional solid blades are not essential. Additionally, employing a grid rotation strategy (such as moving-grid, sliding-grid, or overset-grid) for turbine rotation is also redundant, significantly reducing the number of iterations required for each time step in the simulation process.[84]

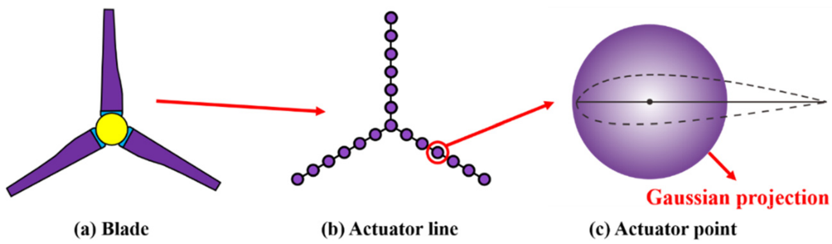

FIG. 2. Schematic diagram of the AL model.

In general, the quantity of APs needs to be approximately 10 greater than the number of grids that run along the radius of the blade. At each time step, these points update the location in accordance with the turbine's predetermined angular velocity. The cross-sectional airfoil's velocity triangle is seen in Fig. 3. U_x and U_r stand for the AP's local axial and tangential velocities, which are linearly interpolated from the velocities of adjacent grids.

The determination of the local attack angle α resembles the BEM theory. Initially, the flow velocity U_x in the domain is assessed through the execution of CFD simulations. Following this, both the local angle of attack α and the local relative inflow velocity U_{rel} are derived from the relationship between the velocity vectors at the blade's cross section (illustrated in Fig. 3). The computation of the body force can be conducted by taking into account the lift and drag characteristics of the airfoil, alongside the Reynolds number relevant to the TST operation. It is important to note that Eq. (4) calculates the local relative inflow velocity U_{rel} at the blade's cross section

$$U_{rel} = \sqrt{U_x^2 + (\omega r - U_r)^2}. \tag{4}$$

The local attack angle α corresponds to the angle formed between the relative inflow velocity U_{rel} and the chord line, which can be determined using the following equation:

$$\alpha = \varphi - \gamma, \tag{5}$$

where γ stands for the pitch angle of the blade section and φ indicates the relationship between the rotor plane and the relative inflow velocity U_{rel}, which is expressed in the following equation:

$$\varphi = \arctan\left(\frac{U_0}{U_r - \omega r}\right). \tag{6}$$

The turbine's lift–drag (L and D) at each actuator point is affected by the local angle of attack α and the local relative inflow velocity U_{rel}, with the relevant expressions provided in the following equations:

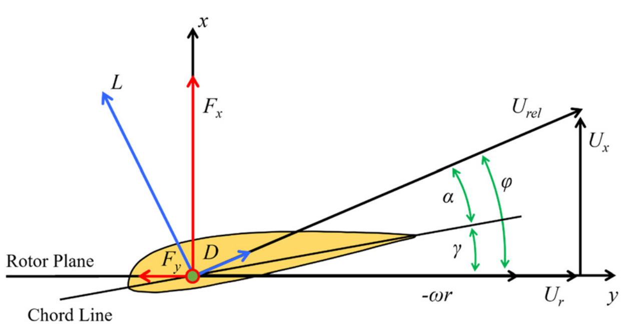

FIG. 3. Velocity vector relationship of the blade cross section.

$$L = \frac{1}{2}\rho w c U_{rel}^2 C_L(\alpha, Re_c), \tag{7}$$

$$D = \frac{1}{2}\rho w c U_{rel}^2 C_D(\alpha, Re_c), \tag{8}$$

where the section width and chord length at the actuator point are represented by w and c, respectively. The lift–drag coefficients (C_L and C_D) are only connected to the local angle of attack α and the local chord length Reynolds number Re_c. The coefficients (C_L and C_D) are utilized to calculate the forcing source term generated by the turbine within the N-S equation. It is important to note that the coefficients are typically obtained through lookup table, XFOIL software, experiments, CFD, etc.[68] Generally, it is necessary to consider the blade tip loss correction $\boldsymbol{F}_{tip}$ in the calculation of the body force, which can be calculated by the following equation:

$$\boldsymbol{F}_{tip} = \frac{2}{\pi}\arccos\left[\exp\left(-\frac{N_{blade}(R - r)}{2r\sin\varphi}\right)\right], \tag{9}$$

where N_{blade} denotes the TST's number of blades and R represents the blade's radius. Additionally, the body force $\boldsymbol{f}_{2D}$ induced by the TST at the actuator point is expressed as

$$\boldsymbol{f}_{2D} = (L\boldsymbol{e}_L + D\boldsymbol{e}_D)\cdot \boldsymbol{F}_{tip}, \tag{10}$$

where $\boldsymbol{e}_L$ and $\boldsymbol{e}_D$ represent the unit vectors corresponding to the lift and drag directions, respectively.

In order to prevent numerical oscillation, it is necessary for the body force $\boldsymbol{f}_{2D}$ induced by the TST to undergo a smooth transition using the Gaussian projection function, as depicted in the following equations:

$$\boldsymbol{f}_{turbine} = \boldsymbol{f}_{2D}\cdot \eta_\varepsilon(r), \tag{11}$$

$$\eta_\varepsilon(r) = \frac{1}{\varepsilon^3\pi^{3/2}}\exp\left(-\frac{r^2}{\varepsilon^2}\right), \tag{12}$$

where $\eta_\varepsilon(r)$ denotes the Gaussian projection function. ε stands for the Gaussian projection width, which is a tuning parameter to adjust the intensity of the body force projection. The value of ε can be calculated using the typical chord length or grid size close to the actuator point. r denotes the distance between the center of the actuator point and the center of the CFD grid affected by the body force.

Many academics worldwide have conducted considerable research on the value of the Gaussian projection width (ε).[85–87] Some academics believe that the ε value should refer to the grid size and propose different suggested values ($\varepsilon \approx 2\Delta x$[85,86] and $\varepsilon \approx 1.76\Delta x$[87]), where Δx represents the smallest grid size in proximity to the actuator points. These suggestions are rooted in the correlation between the numerical outcomes of the turbine and the Gaussian projection width to determine a appropriate ε value. Conversely, other scholars contend that when the grid size adjacent to the blade is significantly smaller than the blade's chord length, the ε value should be influenced by the characteristic chord length of the blade. The characteristic chord length corresponds to the 70%–75% radius position of the blade, which encompasses the primary operating range of the blade, with a proposed value of $\varepsilon \approx c_{70\%\sim75\%}/4$.[88] The earlier works are thoroughly referenced throughout the research for this paper. To account for both the minimum grid size and the characteristic chord length, it is recommended that the Gaussian projection width be taken as the larger of the two, as presented in the following equation:

$$\varepsilon = \max\left[2\Delta x, \frac{c_{70\%\sim75\%}}{4}\right]. \tag{13}$$

In summary, the N-S equation needs to be modified to incorporate the body force source term to accurately represent the TST effect while utilizing the AL code. The procedure for solving the AL code can be outlined as follows:

(1) Acquire the actuator point locations for each parallel zone of the flow field.
(2) Calculating the U_{rel} of each actuator point [Eq. (4)].
(3) Calculating the body force at all actuator points [Eqs. (9) and (10)].
(4) Smooth distribution of the body force by Gaussian projection [Eqs. (11) and (12)].
(5) Solving the flow field [Eqs. (1) and (2)].
(6) Rotating the blades to update the actuator point locations.
(7) Repeat the above process in the following time steps to obtain a transient solution.

C. Vortex identification

The vortex's structure and distribution in turbulent flows have a direct impact on the turbine's wake dynamic characteristics.[89] In order to better comprehend the turbine wake dynamics, it is crucial to identify the vortex structure. However, there is still no agreed-upon solution for identifying vortex structure, and there are divergent viewpoints on the precise definition of vortex. Currently, three generations of vortex identification methods have been developed,[90] as shown in Fig. 4.

The initial method for vortex identification, known as the first-generation approach, relies on Vorticity, which represents the curvature of the velocity vector and is determined by Eq. (14). Investigations[89–91] conducted previously have revealed that vorticity and vortex represent distinct concepts. Despite the fact that they are

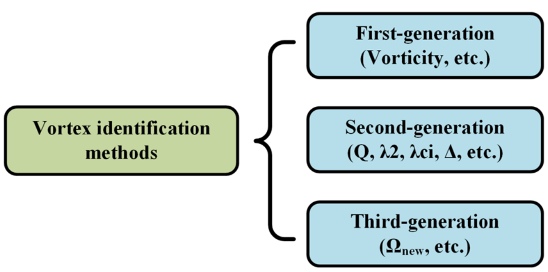

FIG. 4. Classification of vortex identification methods.

both vectors, their magnitude and directional attributes are dissimilar, making them incomparable and non-confusable. Furthermore, numerous researchers[13,61,92] have identified that the first-generation vortex method (including vorticity) is insufficient for accurately characterizing flow field vortex structures, as it can merely outline a general vortex surface profile without adequately representing strong and weak vortices, vortex shedding, or the evolution of vortex structures

$$\text{Vorticity} = \nabla \times \boldsymbol{U}. \tag{14}$$

Owing to the unsatisfactory outcomes from the first-generation vortex method (such as Vorticity), researchers have subsequently developed the second-generation vortex method (including Q, λ_2, λ_{ci}, and Δ), which all acknowledge to some extent that vorticity cannot be utilized to describe the intensity of vortex. The Q and λ_2 methods can be seen as enhancements over the first-generation approach, whereas the λ_{ci} and Δ methods are based on the influence of the velocity gradient tensor ($\nabla \boldsymbol{U}$) on the local instantaneous streamline.[93] Among these four methods, the Q criterion stands out as the most traditional, having been initially introduced by Hunt *et al.*,[94] which employs the second Galilean invariant ($Q > 0$) of the velocity gradient tensor ($\nabla \boldsymbol{U}$) to define the vortex structure. The calculation for the Q criterion is provided in the following equation:

$$Q = \frac{1}{2}\left(\boldsymbol{B}_F^2 - \boldsymbol{A}_F^2\right), \tag{15}$$

where the variables $\boldsymbol{A}$ and $\boldsymbol{B}$ stand for the velocity gradient tensor's symmetric and antisymmetric components, respectively.

The effectiveness of vortex identification is directly impacted by the selection of thresholds established in the last two generations of vortex methods, which is challenging. The threshold choice is irregular, and it is unclear what threshold actually means physically. Moreover, for a clear representation of the wake structure, it is necessary to manually adjust the threshold multiple times across different cases, mesh configurations, and even within the same scenario at various times. In actual flow fields, vortex structures often exhibit varying intensities, and the previous two generations of vortex methods struggle to capture both strong and weak vortices simultaneously. When the chosen threshold is set to a higher value, it results in the exclusion of weak vortexes while only capturing strong vortexes. On the contrary, when the chosen threshold is set to a small value, it becomes possible to detect weak vortexes, but at the cost of reduced clarity in identifying strong vortexes.

In order to address the issues with the earlier two generations of vortex methods, Liu *et al.*[95] innovatively advanced the third-generation vortex techniques, among which the most common method is the Ω_{new} method. The calculation formula for Ω_{new} is presented in Eq. (16). The Ω_{new} value spans from 0 to 1, which can be interpreted as the vorticity concentration. Ω_{new} denotes the rigidity of fluid motion. Specifically, when $\Omega_{new} = 1$, this signifies that the fluid behaves like a rigid body in rotation. If $\Omega_{new} > 0.5$, it signifies that the antisymmetric tensor ($\boldsymbol{B}$) takes precedence over the symmetric tensor ($\boldsymbol{A}$):

$$\Omega_{new} = \frac{\boldsymbol{B}_F^2}{\boldsymbol{A}_F^2 + \boldsymbol{B}_F^2 + \xi}, \tag{16}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ represent the same variables as those defined in Eq. (15). The variable ξ represents a positive value of negligible magnitude,

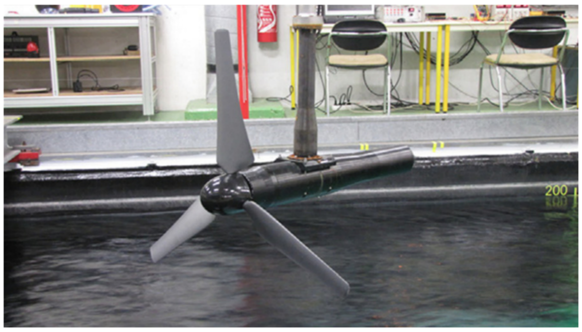

FIG. 5. The schematic diagram of the IFREMER turbine [Reproduced with permission from Mycek *et al.*, Renewable Energy **66**, 729–746 (2014). Copyright 2014 Elsevier].[99]

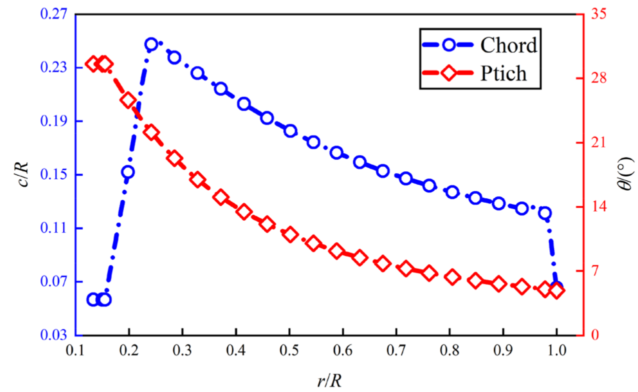

FIG. 6. Radial variation of chord length and pitch angle of the IFREMER turbine.

which serves the purpose of avoiding division by zero. The ξ value is calculated using the following Eq. (17), which avoids the influence of artificially given ξ value on the vortex structure

$$\xi = 0.001 \times \left(B_{\mathrm{F}}^{2} - A_{\mathrm{F}}^{2}\right)_{\max}. \tag{17}$$

In conclusion, the third-generation (Ω_{new}) vortex technique provides several advantages compared to the earlier two generations of vortex methodologies:

(1) a well-defined physical interpretation and straightforward programming implementation;
(2) capable of capturing both strong and weak vortexes concurrently;[96,97]
(3) value normalization facilitates flow control application;[98]
(4) the threshold selection is fixed.[95,96]

In practical applications, for the third-generation (Ω_{new}) vortex method, some scholars[95,96] proposed an initial fixed recommendation threshold ($\Omega_{new} = 0.52$) to identify vortex structures within the flow field. However, this method is still in the development and exploration stages, and its universality has not yet been confirmed across all flow scenarios, despite the fact that it has been used by relevant scholars in a few fields of ocean engineering.[96] Therefore, this paper integrates the third-generation (Ω_{new}) vortex technique into OpenFOAM and successfully applies it to the TST wake visualization, assessing its effectiveness in investigating the TST wake effects.

III. NUMERICAL VALIDATION

The experimental results of the IFREMER turbine[99] were selected to validate the reliability of the AL-DLES coupling method. As shown in Fig. 5, the IFREMER turbine features a three-blade configuration, with the blade section utilizing the NACA63418 airfoil. Figure 6 depicts the chord length and pitch angle distribution at various sections along the radial direction. The diameter of the TST is 0.7 m, and the optimal speed ratio is 3.67. This experiment was conducted in the towing tank of IFREMER in France, where the force characteristics and wake field of the turbine were measured. Furthermore, details of the experiment can be found in the literature.[99]

Figure 7 illustrates the computational domain of the numerical simulation, designed based on the dimensions of the towing tank used in the experiment, with lengths, widths, and heights of 15*D*, 6*D*, and 4*D*, respectively, where *D* represents the diameter of the turbine. The turbine is positioned along the central axis of the calculation domain, with a distance of 3*D* between the turbine's rotating disk and the inlet boundary. The inlet boundary employs a Dirichlet boundary condition, with a flow velocity of 0.8 m/s and a zero pressure gradient. The outlet boundary condition allows for free outflow with a zero pressure gradient, while the top boundary uses a zero-gradient condition. The bottom boundary is defined as a slip wall, and symmetrical boundary conditions are applied to the left and right sides.

The advantage of the AL-DLES coupling method is that the Cartesian grid is used to disperse the entire calculation domain, eliminating the need to solve the flow in the boundary layer of the blade surface, which significantly enhances computational efficiency. The primary advantages of the Cartesian grid include the following: (1) a simple grid structure, (2) high quality, and (3) a user-friendly solver interface. The mesh division of the calculation domain is illustrated in Fig. 8, employing a three-layer step-by-step refinement scheme that consists of the background mesh (mesh region 1), the transition mesh (mesh region 2), and the refined mesh (mesh region 3). In the *x*, *y*, and *z* directions, the background grid size is 0.056 m, while the minimum grid size after refinement is 0.014 m. This refinement scheme helps reduce the dissipation of the wake vortex and is more conducive to studying the turbine wake effect.[68]

To compare with the experimental results, the hydrodynamic characteristics of the turbine under different tip speed ratios (*TSR*s) are calculated. The *TSR* is defined as follows:

$$TSR = \frac{\omega R}{U_0}, \tag{18}$$

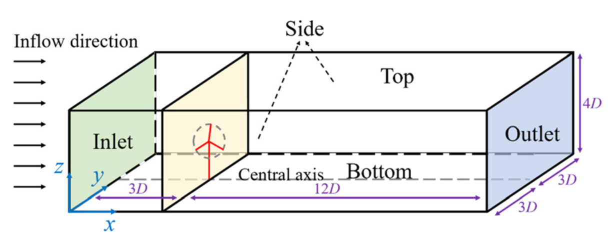

FIG. 7. Calculation domain for the numerical simulation of the IFREMER turbine.

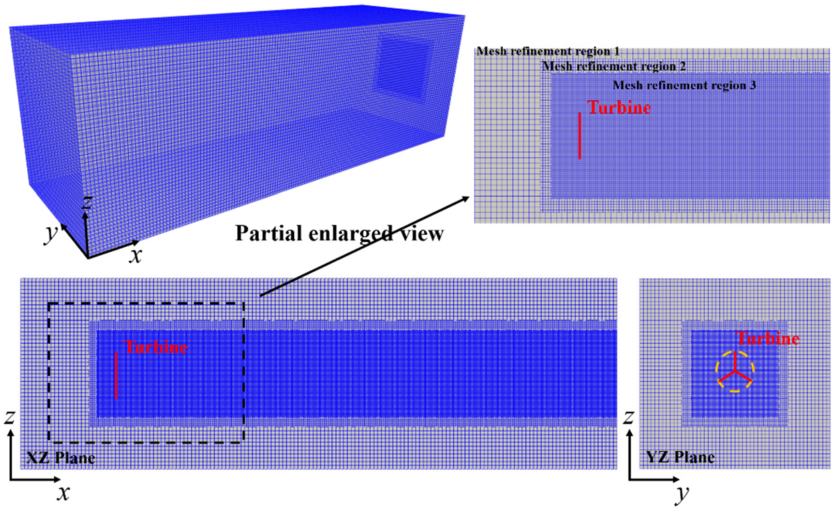

FIG. 8. Mesh distribution in the calculation domain.

where U_0 represents the inflow velocity, ω denotes the turbine's angular velocity, and R signifies the turbine's radius.

To guarantee the stability of the numerical calculations for the explicit scheme, it is necessary for the time step to satisfy the Courant–Friedrichs–Lewy (CFL) stability criterion. This requirement is expressed in Eq. (19), which indicates that the movement of fluid particles during a time step must be less than a specified minimum grid length

$$\text{Courant} = \frac{U\Delta t}{\Delta x} \leq 1, \tag{19}$$

where U denotes the flow velocity and Δx indicates the smallest grid dimension. To ensure proper performance of the turbine, it is essential to comply with Eq. (19) in both the direction of fluid flow (Courant number associated with flow velocity) and the direction of blade rotation (Courant number related to the blade-tip velocity). Consequently, the time step for the numerical simulation is determined based on the duration required for the turbine to complete a rotation of 1°. In addition, the total calculation time needs to ensure that the flow field undergoes 3–4 cycles. Previous investigations conducted by our team[68,84,100] have revealed that if the total calculation time allows for only 1–2 cycles, the wake development of the turbine will be insufficient at high-speed ratios. The final evaluation confirms that this configuration effectively captures the dynamics of the blade vortex system (including both tip and root vortices) and the downstream development process of the vortex.[68]

In terms of validating the hydrodynamic performance of the TST, the hydrodynamic load coefficient is defined as follows:

$$C_P = \frac{P}{0.5\rho A U^3}, \tag{20}$$

$$C_T = \frac{T}{0.5\rho A U^2}, \tag{21}$$

where C_P and C_T represent the turbine's power coefficient and thrust coefficient, respectively. P and T represent the output power and axial thrust of the turbine, respectively. A is the swept area of the turbine. Figure 9 compares the hydrodynamic load coefficients under different tip speed ratios ($1 < TSR < 8$), revealing that:

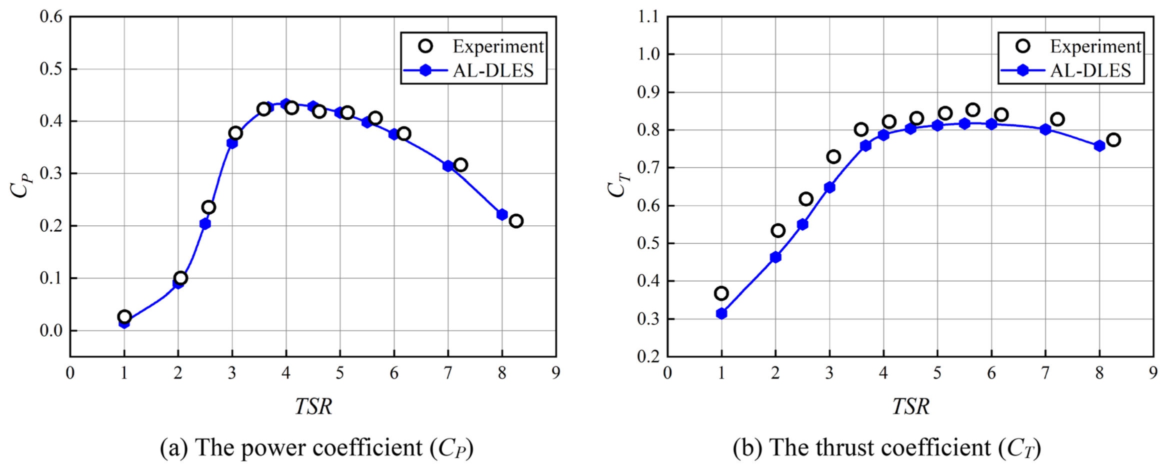

FIG. 9. Hydrodynamic load coefficients of the IFREMER turbine at different *TSR*s.

(1) As the *TSR* increases, the power coefficient of the turbine initially increases and then decreases. When the *TSR* is relatively low, the cross-sectional airfoil of the blade is at a large angle of attack, resulting in flow separation on the blade surface. At this time, the region near the middle and root of the blade is in a stall state with a high drag coefficient, resulting in a significant decrease in the output power of the turbine. When the *TSR* is relatively high, the cross-sectional airfoil of the blade is in a small angle of attack, and the lift coefficient is low, so the power coefficient also begins to drop sharply.
(2) The thrust coefficient of the turbine initially increases but then reduces slightly, peaking at $TSR = 5.5$.
(3) The numerical results are generally consistent with the experimental results. The calculation error of the power coefficient is within 3% near the optimal speed ratio ($TSR = 3.67$). The calculation error of the thrust coefficient is around 5%–10% in all speed ratios ($1 < TSR < 8$). At the same time, the numerical results for the thrust coefficient are all smaller than the experimental results. The phenomenon that the thrust coefficient calculated based on the actuator line model is too small has been reported in many literatures.[53,85,87,101–106] In general, the error of the hydrodynamic load coefficient is within an acceptable range, which validates the reliability of the AL-DLES coupling method in predicting turbine's hydrodynamic loads.

In terms of validating the wake velocity distribution of the TST, selecting the optimal speed ratio case ($TSR = 3.67$), the turbine wake velocity distribution in the calculation stability stage is chosen for comparison. It can be found from Fig. 10 that:

(1) At different flow positions of the wake, the wake width calculated by the AL-DLES method is consistent with the experimental result and slightly larger than the diameter of the turbine, approximately $y = \pm 0.6D$.
(2) The wake velocity deficit is relatively large in the region immediately behind the blade ($-0.5D < y < 0.5D$). The maximum wake velocity deficit captured based on the AL-DLES method is approximately located at the position ($y = \pm 0.2D$). On the contrary, the position of the maximum wake velocity deficit in the experimental results is near the direct rear of the nacelle ($y = 0D$). Across the wake cross sections considered, the AL-DLES method captured the peak of the wake velocity deficit that is close to the experimental results.
(3) The experiments show a V-shaped distribution on all wake cross sections, which is caused by the absorption of incoming flow energy by the turbine. By contrast, the wake shape simulated by the AL-DLES method displays a significant deviation in the near-wake region ($x < 6D$), leading to a W-shaped distribution. Nevertheless, as the flow distance increases, a V-shaped distribution trend appears in the far-wake region ($x > 7D$), which is consistent with the experimental results.
(4) In the near-wake region ($x < 6D$) directly behind the nacelle, the deviation of the AL-DLES method results from the reflected blade rotational effect in the AL model, as it is not possible to simulate the nacelle effect. This results in a significant velocity-lifting phenomenon at the rear of the nacelle.
(5) When the wake further develops into the far-wake region ($x > 7D$), the AL-DLES method results are in good agreement with experimental findings. The difference in wake velocity behind the nacelle decreased gradually as the flow distance increased. The influence of the nacelle on the wake velocity distribution is almost completely eliminated after $x = 8D$.

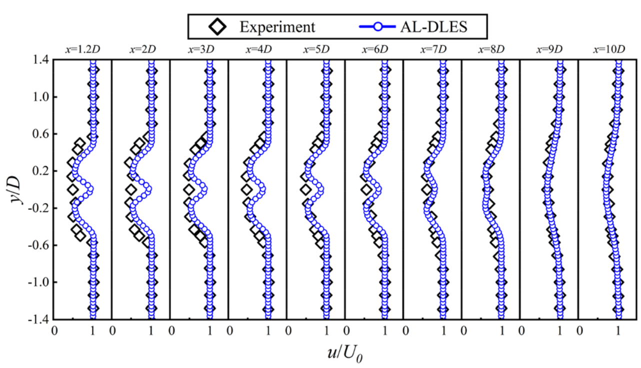

FIG. 10. Wake velocity distribution of the IFREMER turbine at different flow-direction positions.

In summary, the error between the numerical results of the AL-DLES coupling method and the experimental results is within the acceptable range, which validates the accuracy of the AL-DLES coupling method in simulating the hydrodynamic performance and wake characteristics of the TST.

IV. RESULTS AND DISCUSSION

A. Case configuration

Figure 11 defines the coordinate system, azimuth angle, and yaw angle used in the numerical calculations and displays a schematic diagram of the turbine wake under yaw-offset conditions. Selecting four typical yaw angles (0°, 15°, 30°, and 45°) and utilizing the newly developed AL-DLES coupling method, the hydrodynamic and wake dynamics of the TST under yaw-offset conditions are investigated in detail, focusing on the following five aspects: (1) the force characteristics of the turbine (power and thrust coefficient, blade root axial bending moment, blade root tangential bending moment, and turbine yaw moment); (2) the force characteristics of typical blade sections (angle of attack); (3) blade radial force characteristics (lift–drag coefficients, spatiotemporal distribution of angle of attack); and (4) wake characteristics (wake velocity contour and vortex structure).

B. Hydrodynamic performance of the tidal stream turbine

First, the force characteristics of the TST under yaw-offset conditions are discussed. Figure 12 shows the variation of the hydrodynamic load coefficient (power and thrust coefficient) of the turbine with azimuth angle under four typical yaw angles. It is evident that:

(1) Under yaw situations, the hydrodynamic load coefficients of the turbine display periodic fluctuations with azimuth angle, which is a phenomenon that cannot be captured in the axial-flow condition. As the yaw angle is progressively increased, the amplitude of the fluctuation experiences an initial rise followed by a subsequent reduction, and the maximum fluctuation amplitude appears at about the 15° yaw angle.

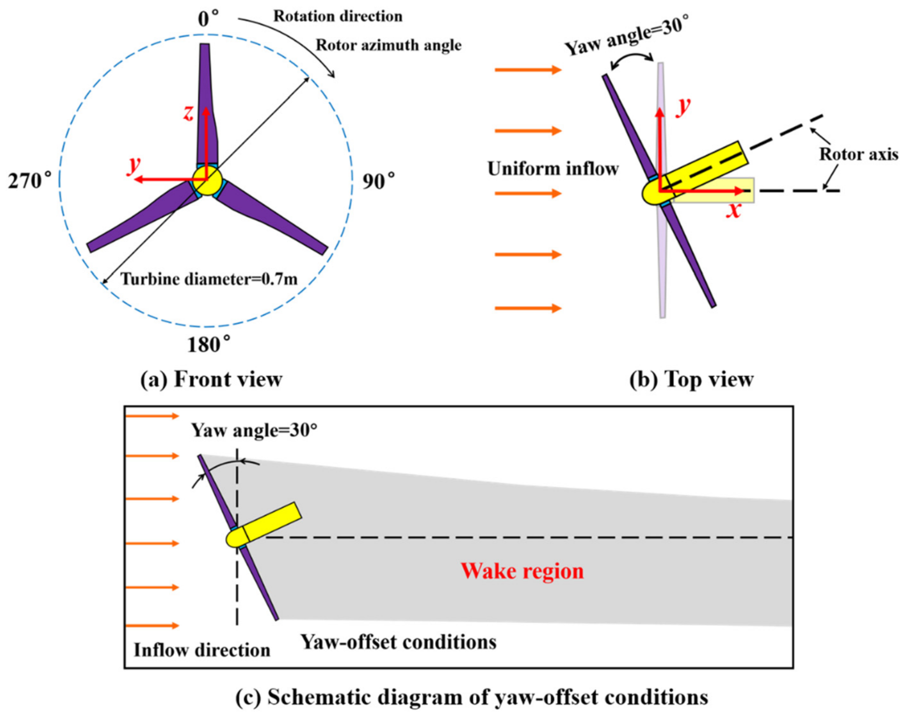

FIG. 11. Definition of the coordinate system for numerical simulation and wake schematic diagram under yaw-offset conditions.

(2) As the yaw angle rises, there is a steady fall in the power and thrust coefficients of the turbine, with the amount of the loss exhibiting a steadily increasing trend. When the turbine is at a minor yaw angle (Yaw = 15°), the power and thrust coefficient only decrease by 8.5% and 5.6%, respectively. At this time, the energy reduction amplitude is not significant. When the yaw angle is between 30° and 45°, the power coefficient decreases by 30%–60%. In addition, the yaw reduces the turbine's output and deflects the wake, reducing the impact on downstream rotors. Therefore, a reasonable application of the yaw angle to upstream rotors could lead to an increase in the overall output power of the tidal array.

When the TST is yawing, the unsteady characteristics of the flow field surrounding the TST are more complicated. The attack angle of each section of the blade varies with the azimuth angle, which will inevitably cause the fluctuation of the blade force. Therefore, regarding the force characteristics of the TST, it is not only necessary to pay attention to the overall hydrodynamic performance (power and thrust) of the turbine, but also to the local load characteristics (root bending moment and yaw moment). The axial bending moment at the blade root (M_n) causes the blade to undergo flapping deformation in the forward and backward directions. The tangential bending moment at the blade root (M_t) causes the blade to undergo swaying deformation along the rotation direction. The turbine yaw moment (M_{yaw}) causes the blade to undergo torsional deformation. The following mainly analyzes three bending moments, which are defined as follows:

$$M_t = \sum_{i=1}^{N_{ap}} (\boldsymbol{f}_i \times \boldsymbol{r}_i) \cdot \boldsymbol{e}_0, \tag{22}$$

$$M_n = \sum_{i=1}^{N_{ap}} (\boldsymbol{f}_i \times \boldsymbol{r}_i) \cdot \boldsymbol{e}_1(\theta), \tag{23}$$

$$M_{yaw} = \sum_{j=1}^{N_b} \sum_{i=1}^{N_{ap}} \boldsymbol{f}_i \times \left(\boldsymbol{r}_i \sin\theta_j\right), \tag{24}$$

where N_{ap} represents the number of actuator points and N_b represents the number of turbine blades. i is the number of actuator points and j is the number of blades. $\boldsymbol{f}_i$ represents the hydrodynamic load acting on the ith actuator point and $\boldsymbol{r}_i$is the displacement vector from the turbine rotation center to the ith actuator point. $\boldsymbol{e}_0$ represents the unit vector along the turbine's rotation axis. $\boldsymbol{e}_1(\theta)$ is the tangential unit vector along the rotation direction when the blade azimuth angle is θ and θ_j is the azimuth angle of the jth blade. Figure 13 shows the relationship between the azimuth angle and the three bending moments (blade root axial bending moment M_n, blade root tangential bending moment M_t, and turbine yaw moment M_{yaw}) of the turbine under four yaw angles. It can be seen that:

(1) When the yaw angle of the TST is 0°, the three bending moments of the turbine approximate a straight line. When the TST has yaw angles, the three bending moments of the turbine fluctuate significantly with the change in azimuth angle. The fluctuation frequency of the axial bending moment (M_n) and tangential bending moment (M_t) at the blade root is close to the turbine rotation frequency. The fluctuation frequency of the turbine yaw torque (M_{yaw}) is twice the turbine rotation frequency. The Lagrangian dynamic LES method adopted in this paper can capture the local transient effects of the non-uniform flow field. When the three bending moments fluctuate significantly, they are also accompanied by high-frequency pulsations

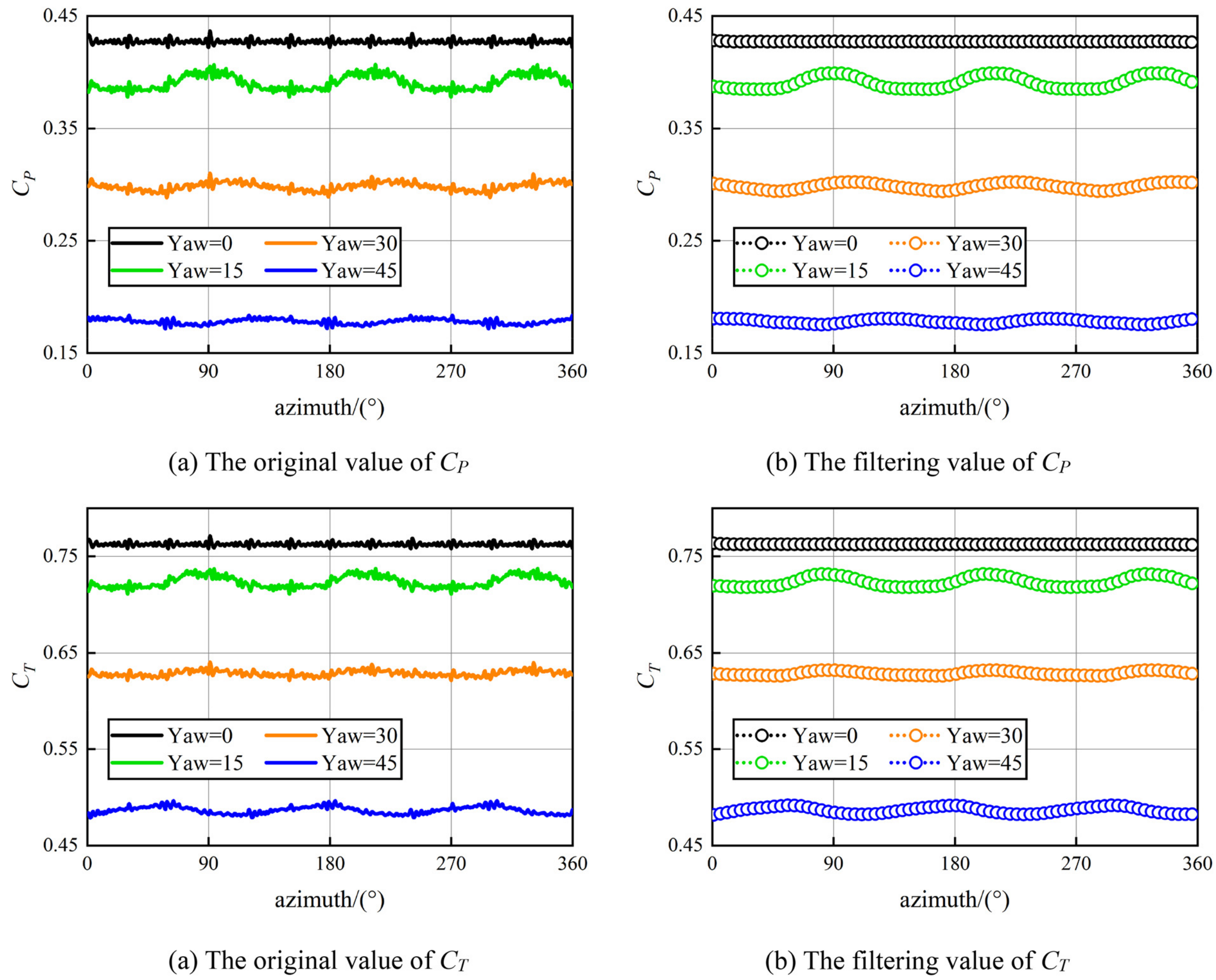

FIG. 12. Variation of the hydrodynamic coefficient of the turbine with azimuth angle under different yaw angles.

caused by local transient effects of the flow field. Therefore, the three bending moment fluctuations caused by the yaw-offset condition will have an adverse effect on the fatigue life of the blades and the joint between the tower and the platform.

(2) At the same yaw angle, the blade root tangential bending moment (M_t) is significantly greater than the blade root axial bending moment (M_n) and the turbine yaw moment (M_{yaw}). Moreover, as the yaw angle increases, the difference between the M_t and the other two moments gradually diminishes.

(3) The fluctuation amplitude of the blade root axial bending moment (M_n) under yaw-offset conditions is the smallest, whereas the fluctuation amplitude of the turbine yaw moment (M_{yaw}) is the largest, indicating that the torsional deformation of the blade caused by the M_{yaw} may be more significant under yaw-offset conditions.

(4) As the yaw angle increases, the mean values of the axial bending moment (M_n) and the tangential bending moment (M_t) at the blade root gradually decrease, while the fluctuation amplitude gradually increases. However, the mean value of the turbine yaw moment (M_{yaw}) remains nearly unchanged, and the fluctuation amplitude also increases.

C. Hydrodynamic performance of typical blade sections

Second, the force characteristics of the blade under yaw-offset conditions are examined. Six typical radial positions of the blade ($r/R = 10\%$, 30%, 50%, 60%, 70%, and 90%) are selected to analyze the impact of the yaw angle on the attack angle α of the typical blade sections. With the presence of a yaw angle, the attack angle α of the blade section changes accordingly, which subsequently affects the force characteristics of the blade. Figure 14 illustrates the variation of the blade's attack angle α with the azimuth angle, and it can be observed that:

(1) When the TST is in an axial-flow condition (Yaw = 0°), the angle of attack α remains almost constant with the azimuth angle. At the blade root position ($0.1R$), the angle of attack α is relatively high, making stalling and flow separation more likely.

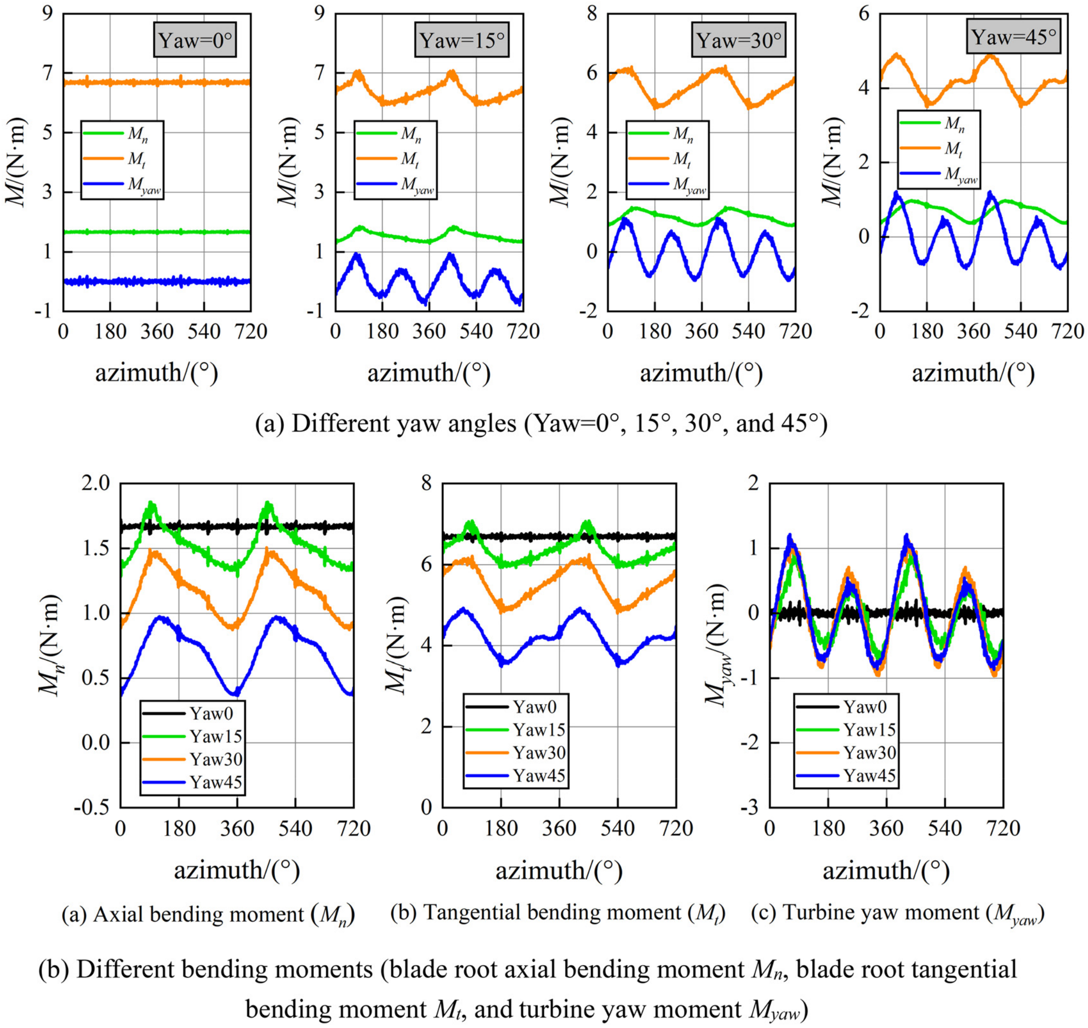

FIG. 13. Variation of the bending moment of the turbine with azimuth angle under different yaw angles.

The angle of attack α from the middle of the blade to the tip ($0.5R$–$0.9R$) is less than 10°, which is close to the optimal angle of attack of the turbine's cross-sectional airfoil. At this time, the lift coefficient exhibits a greater magnitude, while the drag coefficient shows a reduced magnitude. This configuration is more favorable for the assimilation of incoming flow energy, and this range ($0.5R$–$0.9R$) is also the blade's primary working range.

(2) Under yaw-offset conditions, the angle of attack α fluctuates periodically with the change of azimuth angle, and the fluctuation frequency is close to the turbine rotation frequency. At the same yaw angle, the fluctuation amplitude of the attack angle gradually decreases from the root to the tip of the blade. At different yaw angles, the variation pattern of the attack angle is similar. The average attack angle near the blade root ($0.1R$–$0.3R$) is close to that without yaw (Yaw = 0°), and the fluctuation amplitude increases as the yaw angle increases. The average attack angle from the middle of the blade to the tip ($0.5R$–$0.9R$) gradually decreases, and the magnitude of the decrease increases, accompanied by high-frequency fluctuations caused by local transient effects in the flow field.

(3) When the yaw angle is relatively small (Yaw = 15°), the fluctuation of the attack angle from the middle of the blade to the tip ($0.5R$–$0.9R$) is small. At this time, all azimuth angles are always close to the critical angle of attack, which is most conducive to energy absorption. When the yaw angle is large (Yaw = 30°–45°), although the average value of the angle of attack is close to the critical angle of attack, the fluctuation amplitude is large. In certain ranges of azimuth, the smaller angle of attack leads to lower lift–drag coefficients. In addition, the larger the yaw angle, the greater the magnitude of the decrease between the minimum attack angles, which is not conducive to energy absorption. Therefore, if the yaw strategy needs to be introduced to improve the power output of the tidal array, it is preferable to employ a small angle yaw.

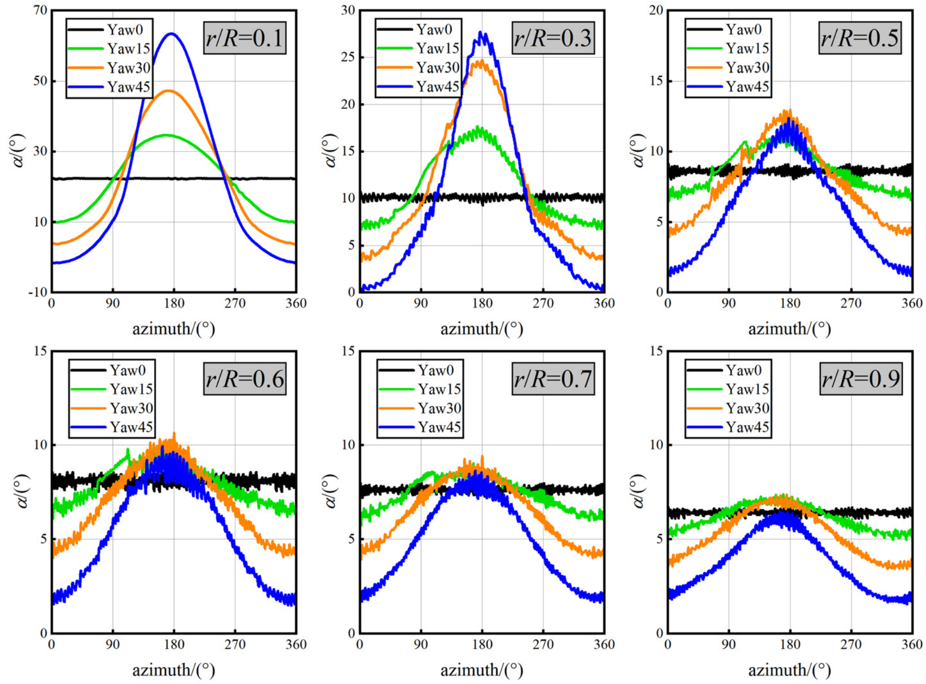

FIG. 14. Variation of attack angle with azimuth angle at six typical sections under different yaw angles.

(4) The peak angle of attack of all sections occurs near the azimuth angle of 180°. Near the blade root ($0.1R$–$0.3R$), the angle of attack of the section near the azimuth angle of 180° is relatively large, which causes a deep stall state for the section airfoil. As the blade develops from root to tip, the angle of attack gradually decreases and approaches the critical angle of attack, yielding the optimal force characteristics.

D. Hydrodynamic performance of blades along radial direction

Subsequently, the radial force characteristics of the blade are analyzed, including lift–drag and angle of attack. Figure 15 shows the radial variation of the lift coefficient *Cl* and drag coefficient *Cd* under four different yaw angles. It can be observed that:

(1) At the same yaw angle, the lift coefficient *Cl* is considerably higher than the drag coefficient *Cd*. The maximum difference range of the lift–drag coefficient occurs in the area near the blade tip, which is the primary working range of the blade.
(2) As the yaw angle increases, the lift coefficient *Cl* gradually decreases, and the magnitude of the decrease accelerates. The variation of drag coefficient *Cd* is relatively small, and the difference between lift–drag coefficient progressively decreases. The decrease in lift coefficient *Cl* is the main reason for the decrease in output power of the turbine under yaw-offset conditions.
(3) Under different yaw angles, the variation trend of the drag coefficient *Cd* is the same, which is first an increase, then a decrease, and finally a hold. Only a high drag coefficient appears near the blade root. At the same time, considering that the area of blade section at the blade root is small, the total drag at the blade root is also small. The variation trend of the lift coefficient *Cl* is similar, with the maximum value range of the lift coefficient appearing in the middle of the blade near the tip.

Figure 16 represents variation of attack angle α along the blade radial direction under varying yaw angles. It is apparent that:

(1) under yaw-offset conditions, due to rotational motion, the blades generate different circumferential linear velocities at different radial positions and azimuth angles, resulting in different attack angles on the blade sections.
(2) Under different yaw angles, the attack angle α followed the same trend along the radial direction, increasing first and then decreasing from the blade root to the tip. The attack angle near the blade root ($0.1R$) is large, resulting in a deep stall phenomenon here. Within the blade's primary work range ($0.3R$–$0.7R$), the change in attack angle of the cross section is stable. At this time, the lift coefficient *Cl* is relatively large, while the drag coefficient *Cd* is small, which absorbs the incoming flow more completely and is most conducive to the turbine's stable operation.
(3) When the yaw angle is relatively small (Yaw = 15°), the attack angle of the blade section is basically the same as that without yaw (Yaw = 0°), maintaining near the critical attack angle. As the yaw angle increases, the attack angle gradually decreases, and the magnitude of the decrease gradually increases.

The yaw-offset condition breaks the rotational symmetry of the hydrodynamic load on turbine blades. By changing the local attack angle of the blades, the turbine's hydrodynamic performance exhibits fluctuating characteristics. Figure 17 shows the spatiotemporal distribution contour of the local attack angle of the blade under varying yaw angles, where the horizontal coordinate represents the azimuth angle

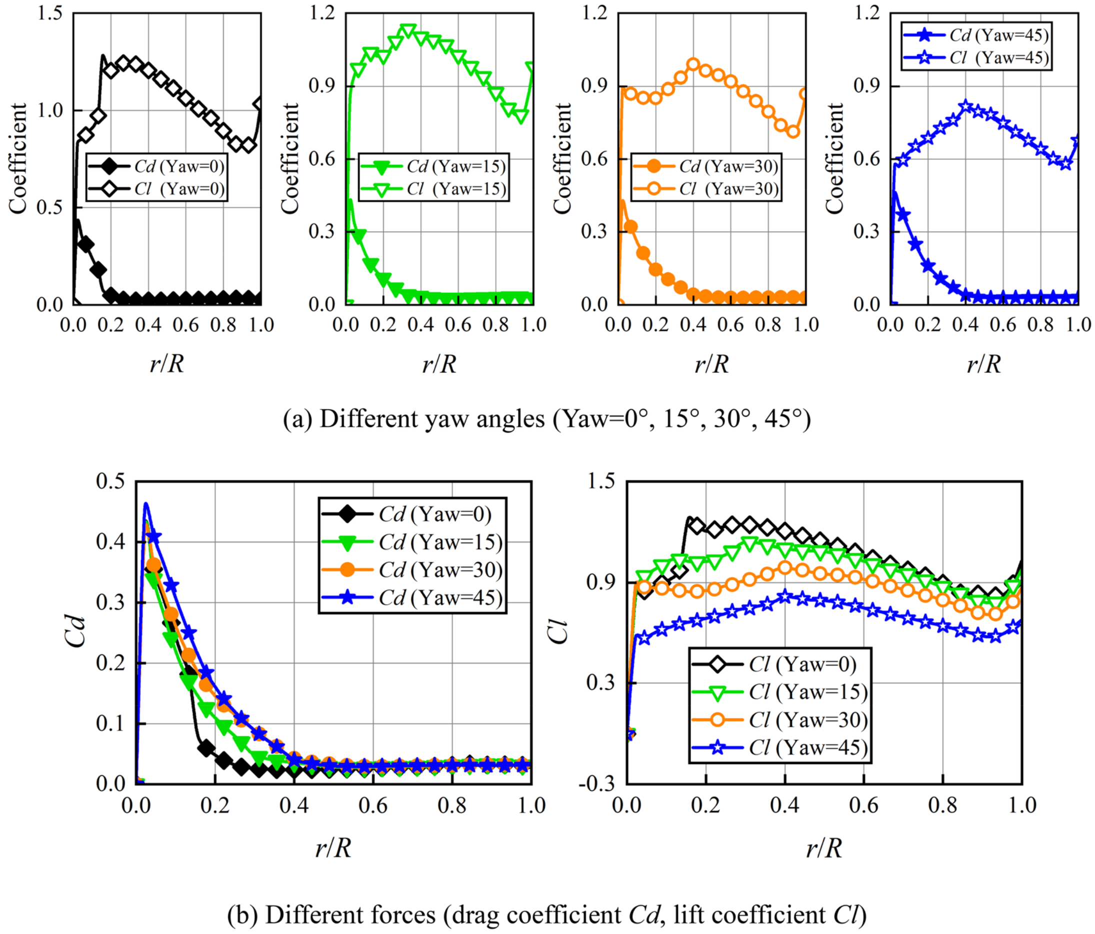

FIG. 15. Variation of lift–drag coefficient along the blade radial direction under different yaw angles.

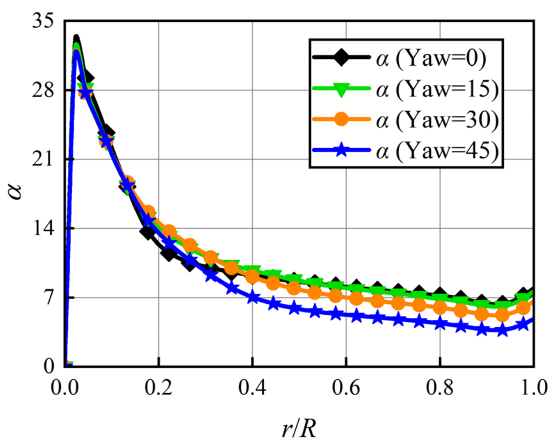

FIG. 16. Variation of attack angle along the blade radial direction under different yaw angles.

and the vertical coordinate represents the relative position of the blade cross section from the root. The contour is colored using the instantaneous fluctuation value of the local attack angle. By comparing with the non-yaw condition (Yaw = 0°), it can be found that:

(1) When the turbine is in a yaw-offset condition, the blade's local attack angle fluctuates periodically. As the yaw angle increases, the difference in local attack angles of the blades is significantly amplified, and the average value of the attack angle decreases. The change in local attack angle will cause a change in the lift–drag coefficient of the blade, which will lead to a fluctuation in the hydrodynamic load of the TST.
(2) As the yaw angle increases, the blade's operating range (the green region in Fig. 17) near the critical attack angle (approximately 8°) gradually decreases. The blade's stall operation range (the red region in Fig. 17) in the range of the large attack angle initially increases and then remains unchanged. However, the change is not obvious for yaw angles between 15° and 45° and is primarily concentrated near the blade root ($r/R < 0.3$). The

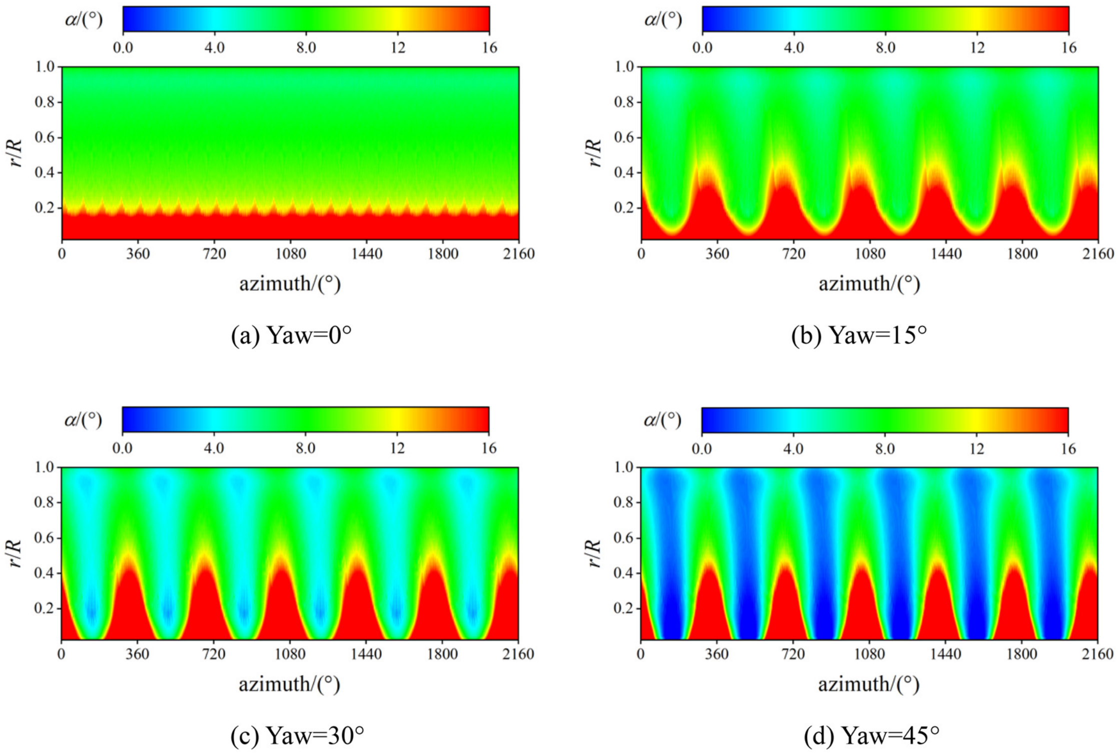

FIG. 17. Spatiotemporal distribution of the local attack angle of the blade under different yaw angles.

blade's operating range (the blue region in Fig. 17) in the range of the small attack angle demonstrates a gradually increasing trend, with the most significant increase occurring at the large yaw angle (Yaw = 45°). Therefore, the primary reason for the decrease in output power due to the yaw angle is the reduction in the operating range of the optimal attack angle, along with an increase in the operating range of the small attack angle.

E. Wake characteristics

Next, the difference of flow field under different yaw angles is analyzed from the perspective of wake velocity contour, which is helpful to understand the flow mechanism of the turbine's yaw wake. Figure 18 shows the velocity contour of the longitudinal section (x–y plane) of the turbine under varying yaw angles, and it can be seen that:

(1) The velocity deficit phenomenon is observed behind the TST at varying yaw angles. As the yaw angle increases, the velocity deficit gradually weakens, making the wake easier to recover, which is caused by the reduction of the thrust coefficient of the turbine along the flow direction.

(2) The yaw-offset condition causes the wake to skew obviously, and the skew angle increases with the yaw angle. Due to the turbine's blocking effect, there is a certain expansion in the initial position of the wake at different yaw angles. As the wake flows downstream, the width of the wake gradually narrows. With the increase of yaw angle, the narrowing of the wake width gradually accelerates.

(3) When Yaw = 0°, the non-physical high-speed region appears behind the nacelle in the velocity contour, which is consistent with the research results of many scholars.[72,84,103–107] This phenomenon is mainly due to the fact that the AL model ignores the nacelle modeling and only considers the rotation effect of blades, resulting in insufficient simulation accuracy of the near-wake behind the nacelle. As the yaw angle increases, the non-physical high-velocity region caused by the lack of nacelle effect gradually weakens, indicating that the issue of inadequate precision in simulating the near-wake by the AL model will be mitigated when subjected to yaw misalignment.

(4) When the turbine's yaw wake develops downstream, the low-speed region of the wake strongly mixes with the surrounding free flow high-speed region, forming a significant meandering phenomenon along the lateral direction. This phenomenon enhances the wake's turbulence and promotes wake recovery. As the yaw angle increases, the wake meandering phenomenon gradually decelerates and weakens. When the turbine is at a large yaw angle (Yaw = 45°), the wake meandering phenomenon has basically disappeared, indicating that the yaw-offset condition can suppress the wake meandering to some extent, so that the lateral influence range of the wake is gradually reduced. Therefore, the reasonable application of the yaw angle to upstream rotors in the tidal array can reduce the lateral interference of the turbine wake using lower output power loss.

Figure 19 represents the velocity contour of the lateral section (y–z plane) of the turbine along the flow direction under varying yaw angles, and it is evident that:

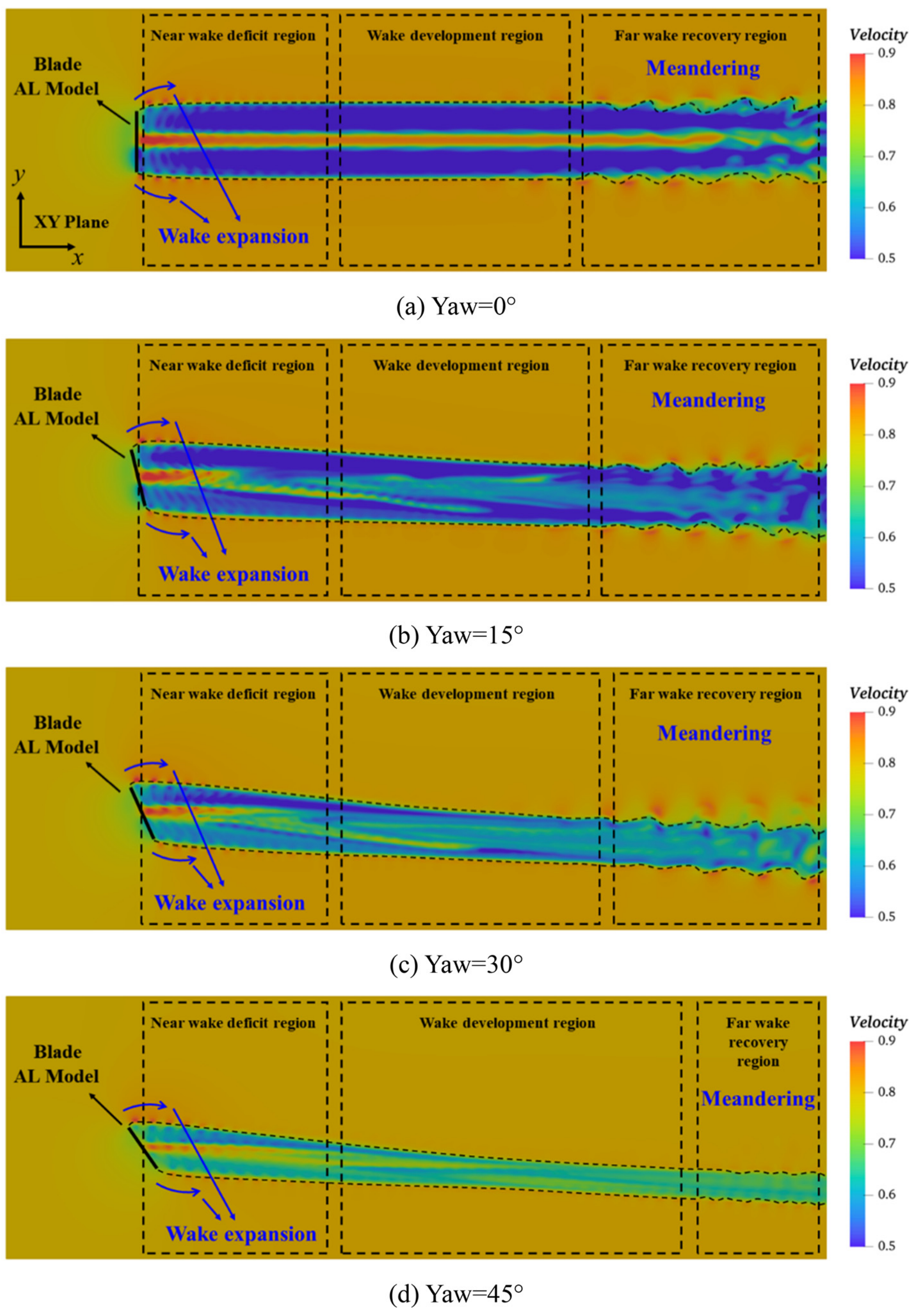

FIG. 18. Wake velocity contour of the turbine along the longitudinal section (x–y plane) under different yaw angles.

(1) When Yaw = 0°, the velocity deficit phenomenon is significant, and the influence range of the velocity deficit is relatively regular, exhibiting a symmetrical circular distribution, indicating that the force on the blade is currently more uniform.

(2) When there are yaw angles, the velocity deficit of the wake cross section presents an elliptical shape in the near-wake region ($x < 3D$). As the wake flows downstream, the shape of the velocity deficit is deformed, which increases the wake's instability. The center of the wake also deviates from its initial position, indicating that the turbine load exhibits fluctuating characteristics at this time. With the increase of yaw angle, the location of wake deformation gradually advances. In addition, due to the mixing effect between the low-speed region of the turbine's wake and the surrounding free flow high-speed region, the wake effect gradually weakens and the radial velocity gradient decreases.

(3) The yaw-offset condition causes the turbine wake to deviate laterally, and the shape of the wake gradually concaves inward due to the uneven pressure distribution in the velocity deficit region.

As the yaw angle increases, the asymmetric wake may even undergo curling deformation and fragmentation. This phenomenon is more significant at a large yaw angle (Yaw = 45°).

F. Wake vortex structure

The turbine wake contains a large number of different-sized vortex structures under yaw-offset conditions. The generation and maintenance of vortex structures will increase the turbulence intensity of the TST wake and fluctuate the TST load. This section introduces the third-generation (Ω_{new}) vortex method into the visualization of turbine vortex structure under yaw-offset conditions for the first time. The universal applicability of the third-generation (Ω_{new}) method to all flow problems has not been validated.[96] Therefore, this section first analyzes the turbine vortex structure under Yaw = 30°. By comparing the differences of three vortex methods (first-generation vorticity, second-generation Q, and third-generation Ω_{new}) in wake vortex visualization, the applicability and benefits of the third-generation Ω_{new} method in the yaw-offset condition are highlighted. Finally, the impact of different yaw angles on the turbine vortex structure is explored based on the latest third-generation Ω_{new} method. Figure 20 illustrates the wake vortex structures captured by three different vortex methods at Yaw = 30° with six different thresholds. Through comparison, it is found that:

(1) The threshold selection of the previous two generations of methods is uncertain and depends on manual experience. Different thresholds have a decisive impact on the identification of wake structures. As the threshold increases, the displayed vortex structure rapidly decreases. Comparatively, the third-generation Ω_{new} method is insensitive to the threshold, and the recommended threshold ($\Omega_{new} = 0.52$)[95] is also applicable to the identification of turbine wake vortex under yaw-offset conditions.
(2) The first two generations of methods are flawed. When the threshold is small, the vortex structure is insensitive to the threshold change. When the threshold is large, there will be the vortex fracture phenomenon, where there should be the weak vortex region (that is, the broken weak vortex region formed by strong compression between the concave blade tip vortex and the blade root vortex). Currently, the previous two generations of methods can only capture strong vortex at small thresholds and weak vortex at large thresholds, making it impossible to simultaneously capture both strong and weak vortex.

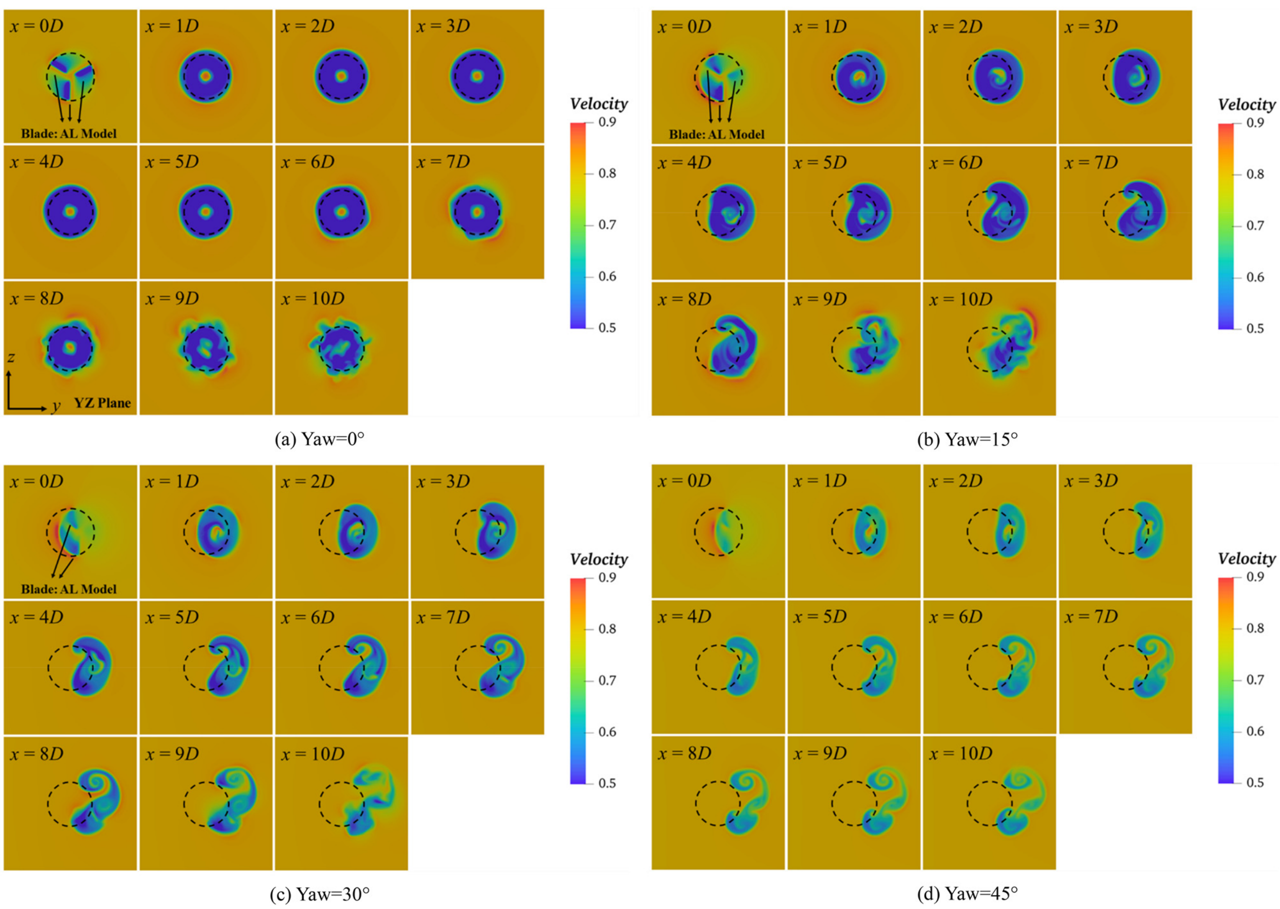

FIG. 19. Wake velocity contour of the turbine along the lateral section (y–z plane) under different yaw angles.

(3) In the position of vortex fracture captured by the previous two generations of methods, the third-generation Ω_{new} method captures the phenomenon of a large-scale vortex breaking into a small-scale weak vortex. This is due to the strong mixing effect between the low-speed region of the turbine wake and the high-speed region of the free flow. The viscous action of the fluid leads to the surrounding fluid being involved in the vortex structure, which causes the vortex structure to break and form a small-scale weak vortex structure. The third-generation Ω_{new} method has the advantage of simultaneously capturing strong and weak vortex, which other vortex identification methods cannot achieve by adjusting the threshold.

(4) The third-generation Ω_{new} method can effectively capture the blade vortex system (tip and root vortex) of the yaw turbine. One side of the blade tip vortex has an obvious depression phenomenon. When the concave tip vortex develops downstream, due to the strong compression effect with the root vortex, large-scale strong vortexes break up and form small-scale weak vortexes. The tip vortex on the other side did not deform, and the vortex spacing gradually widened as it propagated downstream. The root vortex system exhibits a considerable level of complexity, as it undergoes fast disruption due to the combined compression of the concave tip vortex and the blade attachment vortex. Consequently, these vortexes merge and form the vortex surface.

Finally, based on the latest third-generation Ω_{new} method, the differences of turbine vortex structure under different yaw angles are explored, and a fixed recommended threshold ($\Omega_{new} = 0.52$) is adopted. It is found in Fig. 21 that:

(1) Under different yaw angles, the shedding of blade vortex systems (tip and root) and the fusion of concave tip vortex and

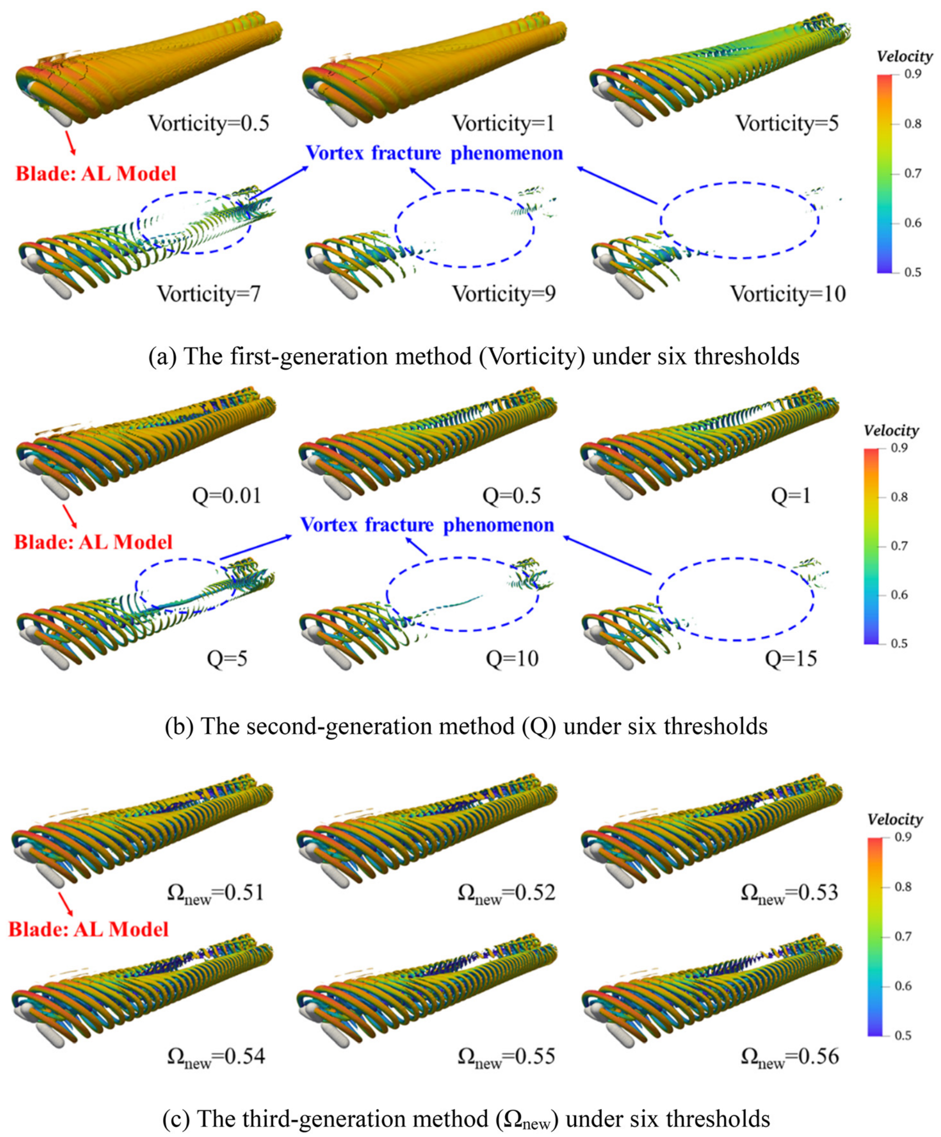

FIG. 20. Under a yaw angle of 30°, three different vortex identification methods visualize the wake vortex structure.

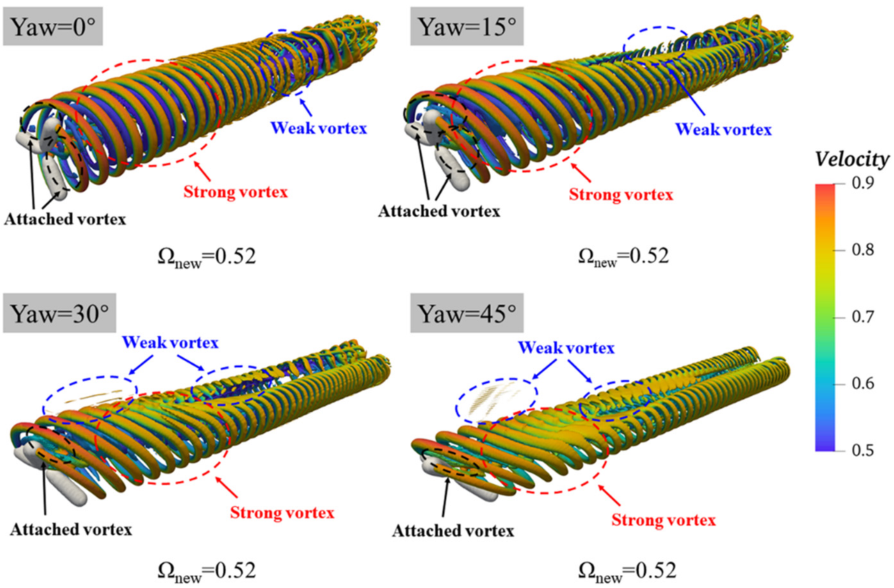

FIG. 21. Under different yaw angles, the third-generation (Ω_{new}) vortex identification method visualizes the wake vortex structure.

root vortex during downstream development can be clearly observed at a fixed threshold.

(2) Due to compression effect, small-scale weak vortex structures formed by the fragmentation of large-scale strong vortex structures can also be identified at the concave position of the blade tip vortex, whereas the previous two generations of methods did not capture this phenomenon.

(3) The yaw-offset conditions cause the wake vortex to tilt, producing a squeezing effect that suppresses wake meandering. This phenomenon is more significant at a large yaw angle (Yaw = 45°). As the yaw angle increases, the concave position of the tip vortex on one side gradually advances and becomes more significant. The tip vortex on the other side did not undergo concave deformation. With the increase of yaw angles, the tip vortex spacing gradually decreased, and the wake vortex width gradually narrowed, thereby promoting the position of vortex mutual interference to advance.

(4) After the incoming flow passes through the rotating blades, the attached vortex is formed on the disk surface of the turbine. As the yaw angle increases, the attached vortex rapidly breaks up and forms the small-scale weak vortex structure. The presence of the attached vortex will result in energy and turbine power losses. During the process of downstream propagation, the attached vortex also interacts with the tip and root vortexes.

V. CONCLUSIONS

Under the background of "carbon peaking and carbon neutrality," the development and utilization of clean energy have become hot topics in both academic and engineering circles. As an important source of clean and renewable energy, tidal stream energy plays an increasingly significant role in the global energy landscape. Given that the cost of tidal stream power generation remains higher than that of thermal power generation, improving the economic viability of tidal stream power is essential for ensuring the sustainable development of the tidal stream industry. Numerical simulation is a crucial tool for studying flow characteristics and turbine power output in large-scale tidal stream fields. Therefore, it is urgent to develop a set of calculation methods suitable for high-accuracy numerical simulations of turbine wakes. This will have practical value and promising development prospects for understanding the wide-ranging spatiotemporal evolution characteristics of TST wakes. Furthermore, the established numerical model will have significant implications for future research.

This study developed a coupling technique known as AL-DLES, in which the rotor of the tidal turbine is represented using the AL model, bypassing the direct resolution of the blade geometry's body-fitted mesh, to gain insights into the anticipated yaw-induced states of the turbine. The integration of a novel Lagrangian dynamic Smagorinsky SGS model enhanced the accuracy of the near-wake simulations, leading to the development of a high-fidelity numerical framework for analyzing the TST wake. The validation of the AL-DLES coupling technique was conducted by referencing benchmark data. Finally, using the AL-DLES coupling method, an investigation was carried out on the hydrodynamic and wake dynamics of the turbine under yaw-offset conditions. Additionally, the latest third-generation Ω_{new} technique was employed in the wake analysis, with a comprehensive comparison of the three vortex identification methods focused on wake visualization, confirming the efficacy and benefits of the third-generation method concerning the turbine wake structure. The conclusions derived from this research are as follows:

(1) At low-speed ratios, the section airfoil of the blade has a large attack angle, which makes it susceptible to stalling and flow separation. Meanwhile, the drag coefficient is high, decreasing the turbine's output power. Comparatively, the blade section airfoil at high-speed ratios has a low attack angle and a low lift coefficient, decreasing output power. Under the optimal speed ratio, the attack angle of different sections varies smoothly and remains close to the critical attack angle, which is most conducive to the turbine's power output.

(2) The yaw-offset condition causes periodic fluctuations in the local attack angle of the blade, with a frequency close to the turbine's rotational frequency. As the yaw angle increases, the difference in local attack angles is significantly amplified. Meanwhile, the operating range of the blades near the critical attack angle gradually decreases, and the stall range at the large attack angle first increases and then remains unchanged, mainly concentrated near the blade root ($r/R < 0.3$). The operating range of the blades at the small attack angle gradually expands. Therefore, the reduction in output power induced by yaw angles can be attributed to the narrowing of the optimal attack angle operating range and the expansion of the small attack angle operating range.

(3) When the yaw angle is small, the load fluctuation on the TST is comparatively minimal, resembling the no-yaw turbine state. Considering that yaw-offset conditions can deflect the wake and thus improve the downstream rotors inflow conditions in the array, the rational application of the small-angle yaw strategy is beneficial to the overall power output of the array.

(4) The previous two generations of vortex identification methods lack clear physical significance, and the selection of threshold plays a crucial role in determining the vortex structure. There are shortcomings that cannot simultaneously capture both strong and weak vortexes. The third-generation Ω_{new} method compensates the above shortcomings.

(5) The yaw-offset condition causes significant deviation in the wake, resulting in an accelerated merging location between the tip and root vortex, which helps to suppress the wake meandering and reduce the influence range of the wake lateral oscillation. As the yaw angle increases, the velocity deficit behind the turbine gradually weakens, facilitating the wake recovery.

(6) After the incoming flow passes through the TST, the attached vortex is formed on the turbine disk surface due to the blade rotation effect. The attached vortex rapidly fragments to form small-scale weak vortex structures as the yaw angle increases. The presence of the attached vortex can reduce the TST's output power. During the process of downstream propagation, the attached vortex also interacts with blade vortex systems (tip and root vortex).

In subsequent research, the intricate interactions of wake disturbances in the large-scale tidal farm will be examined in greater detail in order to enhance the total output power, utilizing the established high-fidelity numerical simulation framework.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Renwei Ji: Conceptualization (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jinhai Zheng:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Resources (equal); Supervision (equal). **Mi-an Xue:** Funding acquisition (equal); Investigation (equal); Methodology (equal); Resources (equal); Supervision (equal). **Ke Sun:** Conceptualization (equal); Investigation (equal); Methodology (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – original draft (equal). **Yonglin Ye:** Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal). **Renqing Zhu:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal); Validation (equal). **E. Fernandez-Rodriguez:** Investigation (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **Yuquan Zhang:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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